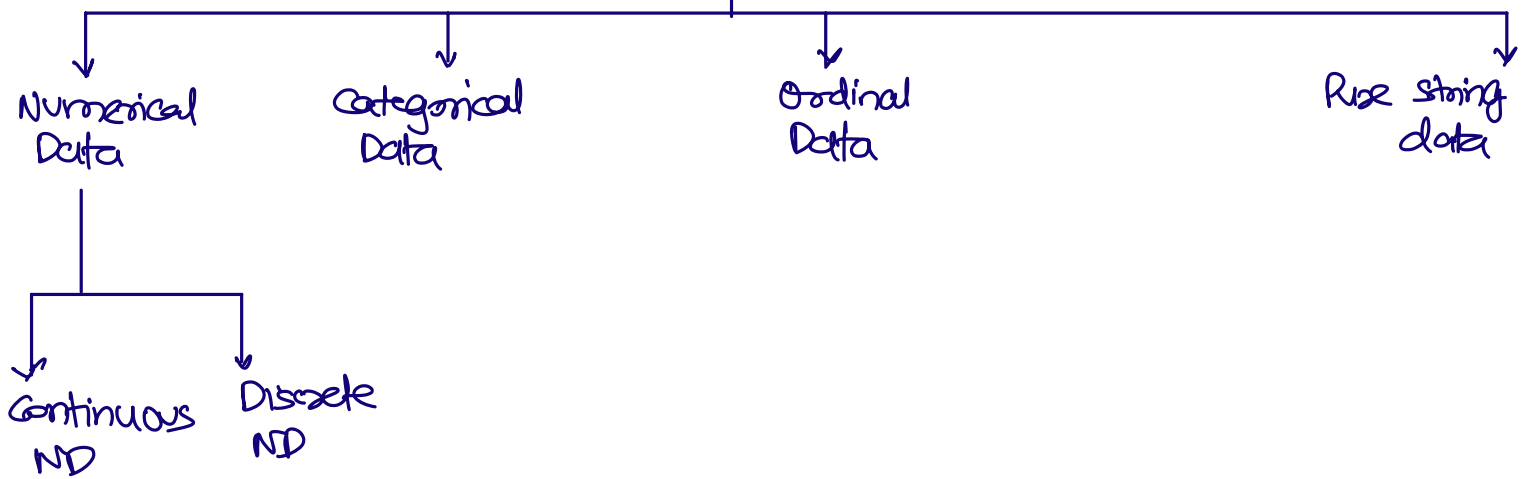


Types of data columns



Statistics

Descriptive stats

- Exploring and understanding the data
- preparing data for inferential stats.

Inferential stats

- Extracting insights
- Creating models.

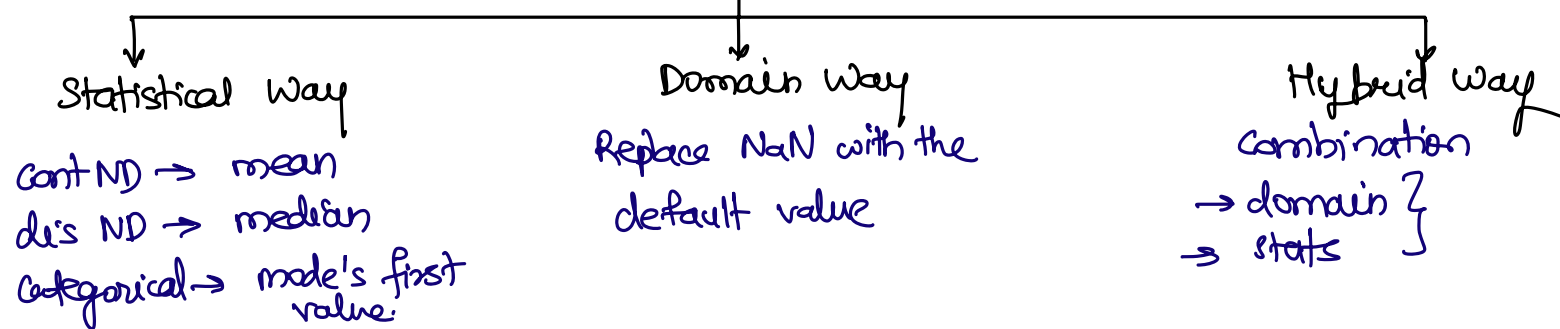
Descriptive Statistics

Data Exploration Phase -or- Exploratory Data Analysis (EDA)

Descriptive Stats is the first step to deal with the SAMPLE. Typical operations that a data analyst, data scientist, business analyst prefer:

- normal ← ————— → skewed
- ① Check the type of **DISTRIBUTION** for each numerical column.
 - ② Identify and deal with **INAPPROPRIATE** data.
 - ③ Identify and deal with **OUTLIERS** —
 - Tukey's method [1.5 IQR Rule]
 - Z score method.
 - ④ Identify and deal with **CATEGORICAL** data. —
 - OHE, Frequency Encoding, Target Encoding
 - ⑤ Identify and deal with **MISSING VALUES**
 - ⑥ Identify and deal with **ORDINAL** data. —
 - Replace category with its RANK
 - ⑦ Discover any **ASSOCIATION RELATIONSHIP PATTERN** —
 - Hypothesis Testing —
 - H_0 — 😞
 - H_a — 😊
 - Statistical Testing Tools
- between two or more VARIABLES (columns) in the given dataset.

Missing value Analysis & Removal (Imputation)

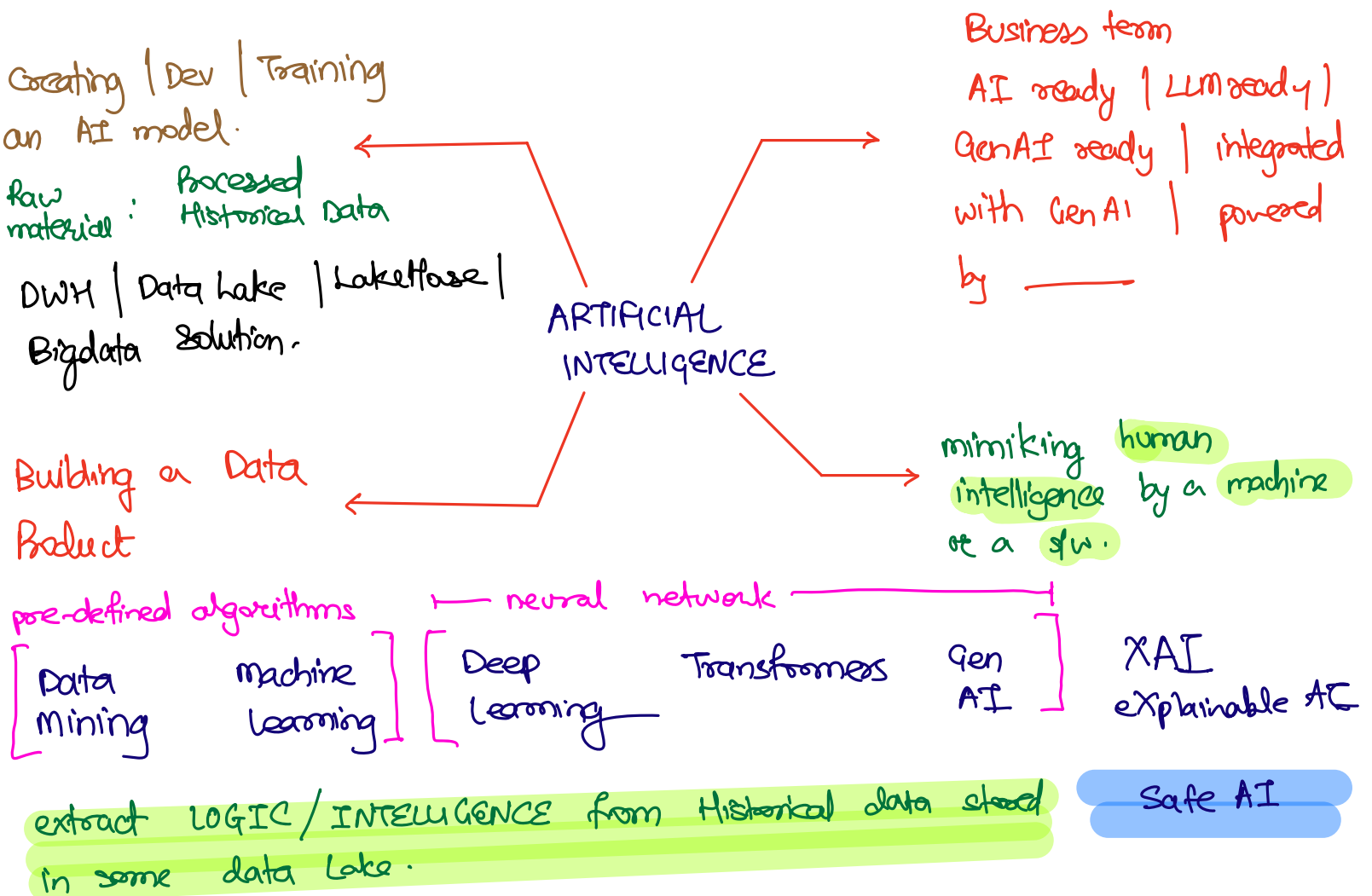


SL [Significance level] → how much errors in prediction or analysis report the business can tolerate.

$$\alpha = 0.05, 0.01, 0.1$$

CL [Confidence level] → how much minimum ^{score} accuracy to be achieved by the data product.

$$CL = 1 - SL.$$



creating an
Artificial
Intelligence
product

} → Data Product

modular Programming
Practices

Here the programmer must be
AWARE of the **LOGIC** to
reach the solution.

Automated Learning
Practices

Here the programmer will make
use of **HISTORICAL** data to
DERIVE LOGIC to reach the
solution.

SQL Engg → DWH
[Preserve
historical
data]

DM
[prepare models
out of the
data]

Bigdata engg
&
statistical
experts → Bigdata
frameworks
eg: Hadoop
Spark

ML
Apache Mahout

Bigdata &
AI engg → Data
Lake

ML, DL
eg: sklearn
tensorflow keras,
pytorch

AI
engg
(Generative
models) → Data
Lake
(unstructured
data)
images, mp3,
video, waveform
...

Generative
models
Adversarial
models

Machine Learning

(It uses data mining algo to derive a model from historical data)

Supervised Learning

Given dataset have both features and label.
(question) (answer)
(input) (output)
(independent variables) (dependent variables)

Given dataset have independent and dependent variables. The goal of supervised learning is to discover the LOGIC to derive the answer.

Unsupervised Learning

Given dataset will have only FEATURES.

Goal: Extract HIDDEN patterns,
Extract HIDDEN categories,
Extract HIDDEN associations

Supervised Learning algorithms

When to use which category of algorithm?

Regression algorithms

Label: Continuous ND.

Classification algorithms

Label: Categorical data
-or-
Discrete ND data.

Model Building process in AI

Assuming the given dataset is **COMPATIBLE** for Inferential stats.

- ① Data must be COMPLETE.
- ② Data must be STRICTLY NUMERIC.

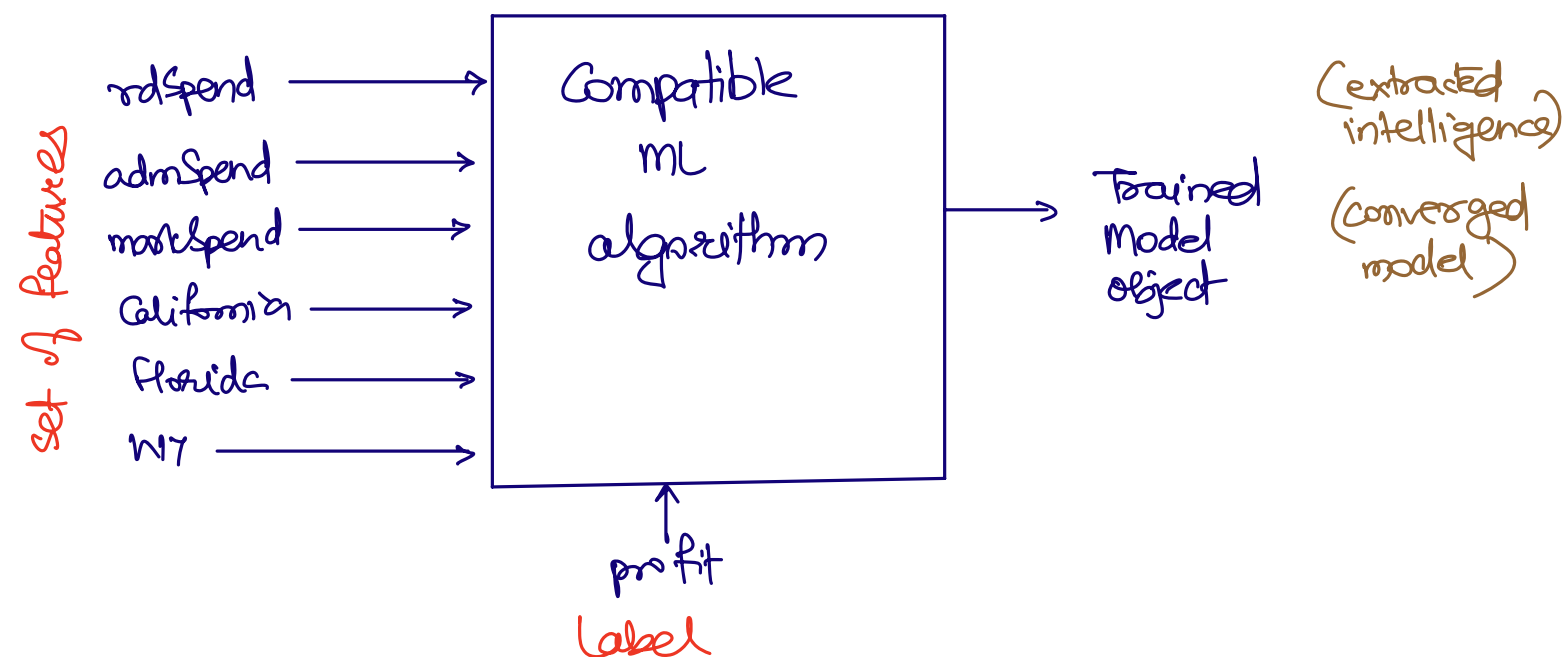
① Separate data as features and label. (Applicable for supervised learning)

② Split the data into two sets

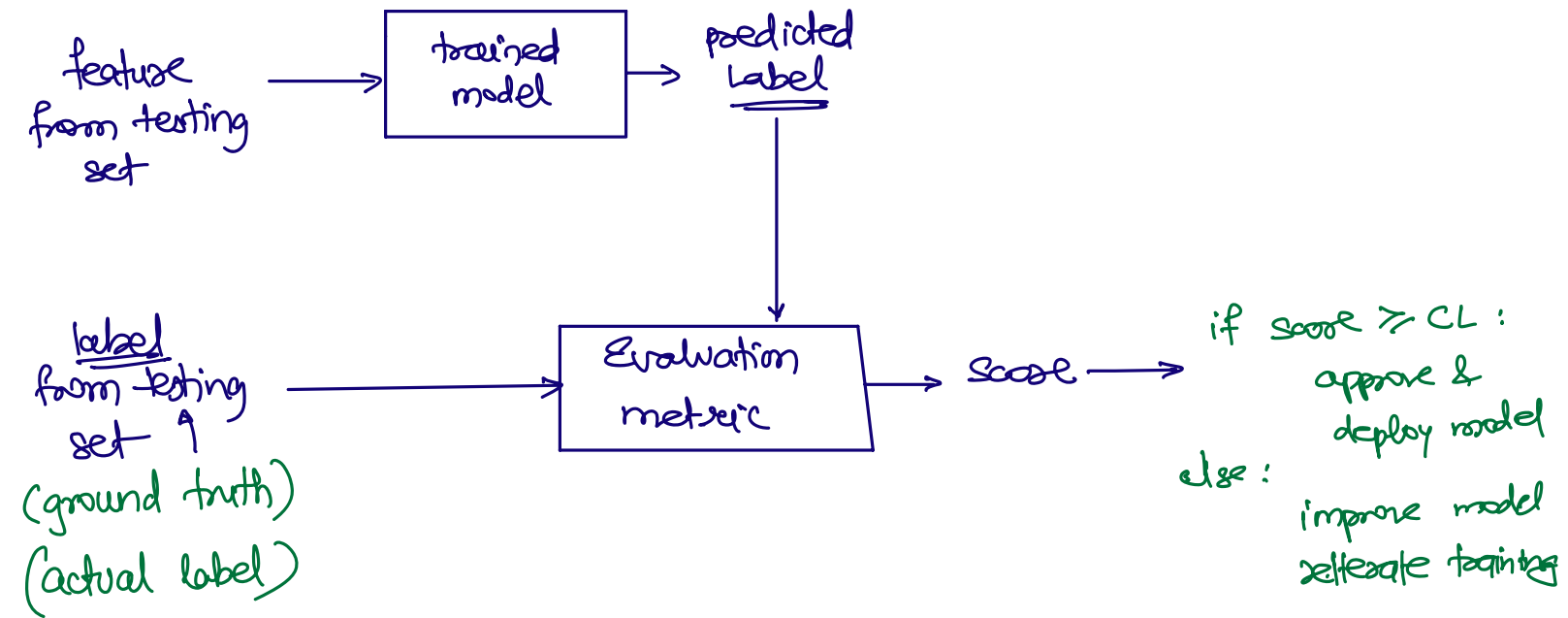
- * Training set → used to TRAIN model [BUILDING | DATA CONVERGENCE]
- * Testing set → used to EVALUATE the model.
(checking the quality of predictions) | model quality

③ Train the model using TRAINING SET and Applicable ML algo.

eg. Create a model that can PREDICT the profit of the company based on company's spending pattern and company's location



④ Check the quality of the model | Model Evaluation.



⑤ Deploy Model.